

0 5 . 1

Contrast the processes of facilitated diffusion and active transport.

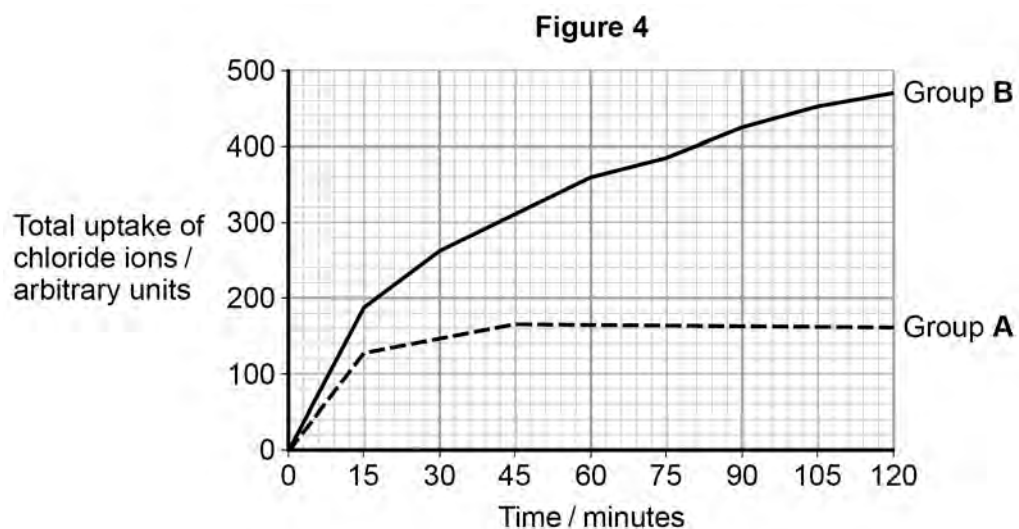
[3 marks]

[Extra space]

Students investigated the uptake of chloride ions in barley plants. They divided the plants into two groups and placed their roots in solutions containing radioactive chloride ions.

- Group **A** plants had a substance that inhibited respiration added to the solution.
- Group **B** plants did not have the substance added to the solution.

The students calculated the total amount of chloride ions absorbed by the plants every 15 minutes. Their results are shown in **Figure 4**.



- 0 5 . 2** Calculate the ratio of the mean **rate** of uptake of chloride ions in the first hour to the **rate** of uptake of chloride ions in the second hour for group **B** plants.

[2 marks]

Ratio = _____ :1

- 0 5 . 3** Explain the results shown in **Figure 4**.

[4 marks]

[Extra space]
